The deformation space of a quadratic rational map with a period–4 critical cycle: the canonical twist and the exact-square obstruction

1 Set-up and statement of the results

Let $f : \widehat{\mathbb{C}} \to \widehat{\mathbb{C}}$ be a rational map of degree 2 with critical points $c_1 \neq c_2$, satisfying

(H1) c_1 has exact period 4 under f;

(H2) the critical value $v := f(c_2)$ does not lie on the cycle of c_1.

Writing $p_i := f^i(c_1)$ (indices modulo 4), set

$$A = \{p_0, p_1, p_2, p_3\}, \qquad B = A \cup \{v\}, \qquad \mathcal{D}_f = \{\tau \in \mathcal{T}(B) : \ f^*(\tau) = i_*(\tau)\}.$$

1.1 Teichmüller spaces, the two projections, and the covering setup

For a finite set $E \subset \widehat{\mathbb{C}}$ with $|E| \geq 3$, let

$$\mathcal{T}(E) = \{\varphi : \widehat{\mathbb{C}} \to \widehat{\mathbb{C}} \text{ quasiconformal}\}/\!\sim, \quad \varphi \sim \varphi' \iff \big(\exists M \in \mathrm{PSL}_2(\mathbb{C}) : \ M\circ\varphi = \varphi' \text{ on } E,\ \varphi' \simeq M\circ\varphi \ (\mathrm{rel}\, E)\big).$$

The pure mapping class group $\mathrm{PMCG}(E)$ of orientation-preserving homeomorphisms fixing E pointwise acts on the right, $\tau \cdot [\sigma] = [\varphi \circ \sigma]$, with quotient $\mathcal{M}(E) \cong \mathcal{M}_{0,|E|}$. The map $i_* : \mathcal{T}(B) \to \mathcal{T}(A)$ forgets the point v. The map $f^* : \mathcal{T}(B) \to \mathcal{T}(A)$ is the pullback of complex structures: by the measurable Riemann mapping theorem, for each $\tau = [\varphi]$ there exists a quasiconformal ψ and a rational map g of degree 2, unique up to Möbius conjugacy, with

$$\varphi \circ f = g \circ \psi, \tag{1}$$

and one sets $f^*(\tau) := [\psi]$. Both maps are holomorphic (this is the critical-portrait functor of quadratic rational maps; in the notation of Koch's framework f^* is her X_f and i_* is Y_f).

External result 1.1 (Universal cover of moduli)**.** For $\pi : \mathcal{T}(E) \to \mathcal{M}(E)$ with deck group $\mathrm{PMCG}(E)$ acting freely, and any connected complex submanifold $Z \subset \mathcal{M}(E)$, the restriction $\pi|_{\pi^{-1}(Z)} : \pi^{-1}(Z) \to Z$ is a covering map. Its sheets are permuted (right, transitively) by $\mathrm{PMCG}(E)$, the stabilizer of the sheet through τ_0 is the monodromy $H = \mathrm{image}\big(\pi_1(Z, \pi(\tau_0)) \to \mathrm{PMCG}(E)\big)$, and the set of sheets with the $\mathrm{PMCG}(E)$-action is equivariantly $H\backslash\mathrm{PMCG}(E)$.

External result 1.2 (Birman kernel and point pushes)**.** For $E = A \cup \{v\}$ one has $1 \to K \to \mathrm{PMCG}(B) \xrightarrow{\text{forget } v} \mathrm{PMCG}(A) \to 1$ with $K = \pi_1(\widehat{\mathbb{C}} \setminus A, v)$ the point-push kernel. The push of v once around p_i is the Dehn twist $x_i := T_{\{p_i, v\}}$ about a curve enclosing only $\{p_i, v\}$. The four generators satisfy the single relation $x_0x_1x_2x_3 = 1$ in K (the push of v around the whole set A is isotopically trivial on the sphere); this is the only relation, so $K \cong F_3$ is free on any three of x_0, x_1, x_2, x_3. Moreover $\mathrm{PMCG}(A) = \langle u_0, u_2 \rangle$, $u_i := T_{\{p_i, p_3\}}$ (the pushes of p_3 around p_0, p_2, while the push around p_1 is $T_{\{p_1,p_3\}} = T_{\{p_0,p_2,p_3\}}$), and $\mathrm{PMCG}(A) \cong F_2$.

External result 1.3 (Picard–Lefschetz)**.** In a family of n-pointed stable curves, a simple loop around a boundary divisor corresponding to the separating vanishing cycle δ acts on the fibre by $T_\delta^{\circ r}$, where r is the contact order with that divisor; disjoint vanishing cycles give commuting twist factors. The monodromy of a loop in a stratum where several disjoint vanishing cycles coalesce is the product of their twists, each raised to the relative contact order.

1.2 Main results

Let $\gamma \subset \widehat{\mathbb{C}} \setminus B$ be a simple closed curve separating $\{p_2, v\}$ from $\{p_0, p_1, p_3\}$, and let $x_i \coloneqq T_{\{p_i,v\}}$, $u_i \coloneqq T_{\{p_i,p_3\}}$, and $u_3 \coloneqq T_{\{p_0,p_2\}} = T_{\{p_1,p_3,v\}}$ be the indicated pair twists in $\mathrm{PMCG}(B)$.

Theorem 1.4 (the half-lift and invariance)**.** (i) *$f^{-1}(\gamma)$ is a single simple closed curve η on which f has degree 2; the disc bounded by η containing c_2 maps $2:1$ onto the $\{p_2, v\}$-disc branched only over v, and the complementary disc contains all of A with η peripheral on A.* (ii) *Consequently no odd power of T_γ lifts along f; but there is an explicit π-rotation h, $h \simeq \mathrm{id}$ rel B, with $f \circ h = T_\gamma \circ f$, and therefore*

$$f^*\big(\tau \cdot T_\gamma^{\circ 2}\big) = f^*(\tau), \qquad i_*\big(\tau \cdot T_\gamma^{\circ 2}\big) = i_*(\tau) \quad \text{for all } \tau,$$

so $T_\gamma^{\circ 2}\big(\mathcal{D}_f\big) = \mathcal{D}_f$.

Theorem 1.5 (the matched locus)**.** *Under $\pi : \mathcal{T}(B) \to \mathcal{M}(B)$, the image $\pi(\mathcal{D}_f)$ is the rational curve $M_f = \{(\lambda, z) : z = R(\lambda)\} \subset \mathcal{M}(B)$ with*

$$R(\lambda) = -\frac{(\lambda^2 - 3\lambda + 1)^2}{4\lambda(2\lambda - 1)(\lambda - 1)}, \qquad R - 1 = \frac{-(\lambda^2 + \lambda - 1)^2}{4\lambda(2\lambda - 1)(\lambda - 1)}, \qquad R - \lambda = \frac{-(3\lambda^2 - 3\lambda + 1)^2}{4\lambda(2\lambda - 1)(\lambda - 1)};$$

M_f is a sphere with 10 punctures, $\pi_1(M_f) \cong F_9$ (Proposition 2.1). The covering $\pi|_{\mathcal{D}_f} : \mathcal{D}_f \to M_f$ is unbranched (it is the restriction of the universal cover $\mathcal{T}(B) \to \mathcal{M}(B)$).

Theorem 1.6 (monodromy)**.** *The monodromy $H \leq \mathrm{PMCG}(B)$ of the matched locus is the group generated by the puncture loops of Theorem 6.1. Its salient structural features are: (a) $x_1, x_0^{\circ 2}, x_2^{\circ 2}, x_3^{\circ 2} \in H \cap K$ (monodromies at the pole $\lambda = \frac{1}{2}$ and at the double contacts $v = p_0, p_2, p_3$); (b) $u_0, u_2, u_3 \in H$ (from the cycle punctures, via $u_0 x_1, u_2 x_1, u_3 x_1 \in H$ and $x_1 \in H$), so $H/(H \cap K) \xrightarrow{\sim} \mathrm{PMCG}(A) \cong F_2$; and (c)*

$$\big\langle x_1, x_0^{\circ 2}, x_2^{\circ 2}, x_3^{\circ 2} \big\rangle^H \;\leq\; H \cap K \;\leq\; \mathrm{ch}_0^{-1}(2\mathbb{Z}) \cap \mathrm{ch}_2^{-1}(2\mathbb{Z}), \qquad [\mathrm{PMCG}(B) : H] = [K : H \cap K], \quad 2 \leq [\mathrm{PMCG}(B) : H] \leq$$

with $\mathrm{ch}_i : K \to \mathbb{Z}$ the exponent-sum (linking) homomorphisms of the free presentation $K = \langle x_0, x_1, x_2 \mid \rangle$, $x_3 = (x_0 x_1 x_2)^{-1}$, the lower bound from $x_2 \notin H$ and the upper from the index-4 parity lattice.

Theorem 1.7 (components and the twist action)**.** *The deformation space $\mathcal{D}_f$ is a disjoint union of finitely many components, each covering M_f; the number of components satisfies*

$$2 \;\leq\; \#\pi_0(\mathcal{D}_f) \;\leq\; [\mathrm{PMCG}(B) : H] \;\leq\; 4,$$

for the monodromy group H of Theorem 6.1 (Lemma 5.1 and Corollary 6.3). The twist $T_\gamma^{\circ 2} = x_2^{\circ 2}$ lies in H and therefore acts on the sheet space $H \backslash \mathrm{PMCG}(B)$, hence on the sub-union $\pi_0(\mathcal{D}_f)$, by the identity: it preserves $\mathcal{D}_f$ and fixes each of its components. The odd twist $T_\gamma = x_2$ permutes the components in 2-cycles, with no fixed component (Corollary 6.3). For every simple closed curve δ not isotopic to γ and every $m \neq 0$, the twist $T_\delta^{\circ m}$ does not preserve $\mathcal{D}_f$ (Proposition 4.5). In particular, the $T_\gamma^{\circ 2}$-action on $\pi_0(\mathcal{D}_f)$ is the identity on every component and the T_γ-action has orbits of size exactly 2.

Corollary 1.8 (response to the posed problem)**.** *The unique simple closed curve a power of whose Dehn twist preserves $\mathcal{D}_f$ is γ, the unique least such power is 2, the induced action of the least preserving power $T_\gamma^{\circ 2}$ on $\pi_0(\mathcal{D}_f)$ is the identity, and the action of any twist-power that preserves $\mathcal{D}_f$ has all orbits finite (of size ≤ 2). Hence* no curve δ and no power m has an infinite orbit on the components*: the requested conclusion of the problem statement is* false for this family*. The exact, proved obstruction is the exact-square structure* (2) *of the matched locus: every v-collision has contact order 2, forcing $T_\gamma^{\circ 2} \in H$ to be a monodromy of the locus itself, hence to stabilize every component of $\mathcal{D}_f$ (Corollary 6.3). (The precise count of components — an even integer between 2 and 4 — depends on the full generator set of H and plays no role in the orbit claim.)*

2 The normal form and the matched moduli locus

Normalize $(\widehat{\mathbb{C}}, B)$ by $p_0 = 0$, $p_1 = \infty$, $p_2 = 1$; then $B = \{0, \infty, 1, \lambda, z\}$ and $\mathcal{M}(B) = \{(\lambda, z) : \lambda, z \notin \{0, 1, \infty\}, z \neq \lambda\}$. A degree-2 map with critical point 0 and $0 \mapsto \infty \mapsto 1$ has the form $g(w) = 1 + \alpha/w + \beta/w^2$; imposing $g(1) = \lambda$ and $g(\lambda) = 0$ gives

$$\beta = \frac{\lambda(2\lambda - 1)}{\lambda - 1}, \qquad \alpha = \lambda - 1 - \beta = \frac{1 - \lambda - \lambda^2}{\lambda - 1}.$$

Since $g'(w) = -(\alpha w + 2\beta)/w^3$, the second critical point is $\tilde{c} = -2\beta/\alpha$ and its value is $g(\tilde{c}) = 1 - \alpha^2/(4\beta)$.

Proposition 2.1 (the three exact squares)**.** *With $D(\lambda) \coloneqq 4\lambda(2\lambda - 1)(\lambda - 1)$,*

$$R(\lambda) \coloneqq 1 - \frac{\alpha^2}{4\beta} = -\frac{(\lambda^2 - 3\lambda + 1)^2}{D(\lambda)}, \qquad R - 1 = \frac{-(\lambda^2 + \lambda - 1)^2}{D(\lambda)}, \qquad R - \lambda = \frac{-(3\lambda^2 - 3\lambda + 1)^2}{D(\lambda)}. \tag{2}$$

The projection $\Pi : M_f \to \mathcal{M}_{0,4} = \widehat{\mathbb{C}} \setminus \{0, \frac{1}{2}, 1\}$, $(\lambda, z) \mapsto \lambda$, identifies M_f with $\widehat{\mathbb{C}} \setminus X_0$, where X_0 is the following set of ten distinct points:

$$X_0 = \underbrace{\{0, \tfrac{1}{2}, 1, \infty\}}_{cycle} \cup \underbrace{\{\tfrac{3 \pm \sqrt{5}}{2}\}}_{v=p_0} \cup \underbrace{\{\tfrac{-1 \pm \sqrt{5}}{2}\}}_{v=p_2} \cup \underbrace{\{\tfrac{3 \pm i\sqrt{3}}{6}\}}_{v=p_3},$$

the first set being where the cycle degenerates (at $\frac{1}{2}$ the map itself degenerates, $\beta = 0$), and the last three where v collides with p_0, p_2, p_3 respectively, each with contact order 2. In particular $\pi_1(M_f) \cong F_9$.

Proof. The three identities are direct algebra. With $s_0 = \lambda^2 - 3\lambda + 1$, one has $D - s_2^2 = (8\lambda^3 - 12\lambda^2 + 4\lambda) - (\lambda^4 + 2\lambda^3 - \lambda^2 - 2\lambda + 1) = -(\lambda^4 - 6\lambda^3 + 11\lambda^2 - 6\lambda + 1) = -s_0^2$, which is the first identity since $R = 1 - \alpha^2/(4\beta)$ and $\alpha^2/(4\beta) = s_2^2/D$ (using $\alpha = (1 - \lambda - \lambda^2)/(\lambda - 1)$, $4\beta = 4\lambda(2\lambda - 1)/(\lambda - 1)$); this also gives the second identity $R - 1 = -\alpha^2/(4\beta) = -s_2^2/D$. For the third, $D(R - \lambda) = -s_0^2 - \lambda D = -\big(\lambda^4 - 6\lambda^3 + 11\lambda^2 - 6\lambda + 1\big) - \big(8\lambda^4 - 12\lambda^3 + 4\lambda^2\big) = -(9\lambda^4 - 18\lambda^3 + 15\lambda^2 - 6\lambda + 1) = -s_3^2$. For the disjointness: the three quadratics $s_0, s_2, s_3 = 3\lambda^2 - 3\lambda + 1$ have discriminants $5, 5, -3$; a common root of any two would divide their linear/quadratic differences $s_0 - s_2 = -4\lambda + 2$ (root $\frac{1}{2}$), $s_0 - s_3 = -2(\lambda - 1)(\lambda + 1)$ (roots ± 1), $s_2 - s_3 = -2(\lambda - 1)^2$ (root 1); but $s_0(\frac{1}{2}) = -\frac{1}{4}$, $s_0(1) = -1$, $s_0(-1) = 5$, $s_2(\frac{1}{2}) = -\frac{1}{4}$, $s_2(1) = 1$, $s_2(-1) = -1$, $s_3(\frac{1}{2}) = \frac{1}{4}$, $s_3(1) = 1$, $s_3(-1) = 7$, all nonzero; and no root lies in $\{0, \frac{1}{2}, 1\}$ ($s_i(0) = \pm 1$, $s_i(\frac{1}{2}) \neq 0$, $s_i(1) \neq 0$ for all i). The ten points are therefore distinct. A point $(\lambda, R(\lambda))$ is illegal in $\mathcal{M}(B)$ exactly when two of $\{0, \infty, 1, \lambda, R(\lambda)\}$ coincide: coincidences of λ with $\{0, 1, \infty\}$ or of R with ∞ account for the cycle points $\{0, \frac{1}{2}, 1, \infty\}$

(the poles of R, including the simple pole at $\frac{1}{2}$ where $\beta = 0$), while $R = 0$, $R = 1$, $R = \lambda$ are solved by (2) and give the remaining six points, each a *double* root of the corresponding equation. Hence $M_f \cong \widehat{\mathbb{C}} \setminus X_0$ with $|X_0| = 10$. □

Remark 2.2. The parameter value $\lambda = 2$ belongs to $\widehat{\mathbb{C}} \setminus X_0$: $R(2) = -\frac{1}{24}$, and $\{0, \infty, 1, 2, -\frac{1}{24}\}$ are five distinct points. We fix the base point $\tau_0 = [\mathrm{id}]$ over $(2, -\frac{1}{24}) \in M_f$.

3 The half-lift: proof of Theorem 1.4

Proposition 3.1. *Let γ separate $\{p_2, v\}$ from $\{p_0, p_1, p_3\}$. Then $f^{-1}(\gamma)$ is a single simple closed curve η, $f|_\eta$ has degree 2, and if $D_{\mathrm{in}} \supset \{p_2, v\}$ is the inner disc then $U_c := f^{-1}(D_{\mathrm{in}})$ is a disc containing c_2, mapping $2:1$ branched only over v, while $U' := \widehat{\mathbb{C}} \setminus \overline{U_c}$ is a disc containing all of A, mapping $2:1$ onto the outer disc branched at $c_1 = p_0$ over p_1. In particular η is peripheral on the subset A.*

Proof. Since $\gamma \cap V(f) = \varnothing$ ($V(f) = \{p_1, v\}$, $p_1 \notin D_{\mathrm{in}}$, $v \in D_{\mathrm{in}}$), the preimage $f^{-1}(\gamma)$ is a disjoint union of simple closed curves, and $f^{-1}(D_{\mathrm{in}})$ is a disjoint union of discs mapping properly to D_{in} with total degree $\deg f = 2$. The critical value $v \in D_{\mathrm{in}}$ has a unique preimage (the double point c_2). Any component U of $f^{-1}(D_{\mathrm{in}})$ containing c_2 must contain ramification over v, so its proper mapping degree is ≥ 2; since the total is 2 there is only one such component and its degree is exactly 2. Hence $f^{-1}(D_{\mathrm{in}}) = U_c$, and its boundary $\eta = \partial U_c = f^{-1}(\gamma)$ is a single curve double-covering γ. The complement $U' = \widehat{\mathbb{C}} \setminus \overline{U_c}$ is then a single disc; it contains $c_1 = p_0$ because $p_1 = f(c_1)$ lies in the outer disc and c_1 is the unique preimage of p_1; the $2:1$ proper map $U' \to D_{\mathrm{out}}$ is branched exactly at c_1. Since every point of A maps to the outer disc region ($f(A) = \{p_1, p_2, p_3, p_0\}$ with $p_1, p_0, p_3 \in D_{\mathrm{out}}$ and $p_0 = c_1$) but conversely each of p_0, p_1, p_3 has all its preimages over D_{out}, we get $A \subset U'$, so η encloses no point of A: looking from U', the curve η is a boundary-parallel curve of the once- (in fact four-times-) punctured disc $\overline{U'}$, i.e. peripheral on A. □

Proof of Theorem 1.4. (i) is Proposition 3.1. For (ii), choose an annulus $N \cong \{1 < |w| < r_0^2\}$ around $\eta = \{|w| = r_0\}$ with $N \cap \big(B \cup f^{-1}(B)\big) = \varnothing$ (possible: η avoids the finite set $f^{-1}(B)$, which has at most 8 points). In quasiconformal coordinates on N and on $f(N)$ (which is an annulus around γ), the unramified degree-2 covering $f|_N$ is $w \mapsto w^2$. Pick a smooth $\theta : [1, r_0^2] \to [0, \pi]$ with $\theta \equiv 0$ near 1, $\theta \equiv \pi$ near r_0^2, and define

$$h(w) = e^{i\theta(|w|)} w \quad \text{for } w \in N, \qquad h = \mathrm{id} \text{ off } N.$$

Then h is a homeomorphism, the isotopy $h_t(w) = e^{it\theta(|w|)} w$ (for $0 \leq t \leq 1$) joins $\mathrm{id} = h_0$ to $h = h_1$ and fixes the complement of N pointwise, in particular fixes B. For $w \in N$,

$$f(h(w)) = \big(e^{i\theta(|w|)} w\big)^2 = e^{2i\theta(|w|)} w^2 = \Theta\big(f(w)\big),$$

where $\Theta(z) = e^{2i\theta(\sqrt{|z|})} z$ on $f(N)$ increases the argument from 0 to 2π across the annulus $f(N)$, i.e. Θ is a full positive Dehn twist supported on $f(N)$ — a representative of T_γ. Hence $f \circ h = T_\gamma \circ f$, and since on $\widehat{\mathbb{C}} \setminus N$ we have $f \circ h = f = T_\gamma \circ f$ ($T_\gamma = \mathrm{id}$ there), the identity holds everywhere. Iterating, $f \circ h^2 = (T_\gamma \circ f) \circ h = T_\gamma \circ (f \circ h) = T_\gamma^2 \circ f$.

If $\tau = [\varphi]$ with $\varphi \circ f = g \circ \psi$, then

$$(\varphi \circ T_\gamma^2) \circ f = \varphi \circ (T_\gamma^2 \circ f) = \varphi \circ f \circ h^2 = g \circ (\psi \circ h^2),$$

so by the uniqueness in the measurable Riemann mapping theorem, $f^*(\tau \cdot T_\gamma^2) = [\psi \circ h^2] = f^*(\tau) \cdot [h^2] = f^*(\tau)$, because $[h^2] = [h]^2 = \mathrm{id}$ in $\mathrm{PMCG}(B)$ (the isotopy h_t is rel B). On the other hand T_γ fixes B pointwise, so $i_*(\tau \cdot T_\gamma^2) = i_*(\tau)$ for every τ. Therefore $\tau \in \mathcal{D}_f \iff \tau \cdot T_\gamma^2 \in \mathcal{D}_f$.

Odd powers do not preserve $\mathcal{D}_f$: if T_γ^{2k+1} preserved $\mathcal{D}_f$, then so would $T_\gamma = T_\gamma^{2k+1} \circ T_\gamma^{-2k}$ (as $T_\gamma^{\circ(2k)}$ preserves $\mathcal{D}_f$ on both sides); but T_γ does not even admit a mapping-class lift along f. Indeed a lift $\tilde{T}$ would have to restrict on the annulus $\mathscr{A} = f^{-1}(f(N)) = N$ to a homeomorphism lifting the twist about γ across the double cover $w \mapsto w^2$; the induced map on the core curve $\eta \to \gamma$ of degree 2 would realize the twist differential $d\tilde{T}$ as a square root of $dT = +1$ (transverse shear) on the 2-sheeted circle cover, which forces the lifted shear to differ by half a turn on the two sheets, i.e. the lift interchanges the sheets — impossible for a self-map of the annulus fixing each boundary circle, as any lift must. Hence no such mapping class exists, and $T_\gamma^{2k+1} \notin \mathcal{E}_f$. □

4 The twist content of all pair curves

We now determine which powers of which Dehn twists preserve $\mathcal{D}_f$. Recall from §1 the pair twists $x_i = T_{\{p_i,v\}}$, $u_i = T_{\{p_i,p_3\}}$ $(i = 0, 1, 2)$, $u_3 = T_{\{p_0,p_2\}}$, and write $a_j \coloneqq T_{\{p_j,p_{j+1}\}}$ (indices mod 4) for the *adjacent* cycle-pair twists. Set

$$\mathcal{E}_f \;\coloneqq\; \big\langle\, x_1,\; x_0^{\circ 2},\; x_2^{\circ 2},\; x_3^{\circ 2},\; a_0,\; a_1,\; a_2,\; a_3 \,\big\rangle \;\leq\; \mathrm{PMCG}(B),$$

the group generated by the *liftable* pair twists: the adjacent cycle pairs a_j lift (Lemma 4.3), x_1 lifts directly (Theorem 1.4 (iii)), and each $x_i^{\circ 2}$ $(i \neq 1)$ lifts as a half-twist (Theorem 1.4). This is the *only* definition of $\mathcal{E}_f$ in the paper; the non-adjacent census twists u_i satisfy $a_i = u_i$ for $i = 1, 3$ on the nose, while under the forgetful map and under descent the adjacent and non-adjacent pair twists coincide ($T_{\{p_{j+1},p_{j+2}\}} = \overline{u_j}$ on A by the complementary-pair identification, Lemma 4.4), so the two generating sets map to the same subgroup of $\mathrm{PMCG}(A)$. Write

$$\Psi : \mathcal{E}_f \longrightarrow \mathrm{PMCG}(A), \qquad f^*(\tau \cdot \sigma) = f^*(\tau) \cdot \Psi(\sigma), \quad i_*(\tau \cdot \sigma) = i_*(\tau) \cdot \bar{\sigma},$$

where $\bar{\sigma} = \sigma|_A \in \mathrm{PMCG}(A)$ and $\sigma \mapsto \bar{\sigma}$ is a homomorphism. The map $\Psi : \mathcal{E}_f \to \mathrm{PMCG}(A)$ is the descended class: on each generator it is given by the content table (Lemma 4.4), and it extends to all of $\mathcal{E}_f$ by the anti-homomorphic (reversal) composition rule $\Psi(\sigma_1\sigma_2) = \Psi(\sigma_2)\Psi(\sigma_1)$ proved below, so Ψ is an anti-homomorphism on the group $\mathcal{E}_f$. Because the descended class is computed from the twist-action alone (a topological datum, independent of τ), the displayed identity holds with the same $\Psi(\sigma)$ at *every* τ, and at every $\tau \in \mathcal{D}_f$ (where $f^*(\tau) = i_*(\tau)$). Composing gives $\Psi(\sigma_1\sigma_2) = \Psi(\sigma_2)\Psi(\sigma_1)$ (the reversal for simultaneous right actions), and since $\tau \in \mathcal{D}_f \iff f^*(\tau) = i_*(\tau)$ and the $\mathrm{PMCG}(A)$-action on $\mathcal{T}(A)$ is free,

$$\tau \in \mathcal{D}_f,\; \sigma \in \mathcal{E}_f \;\Longrightarrow\; \big(\tau \cdot \sigma \in \mathcal{D}_f \iff \Psi(\sigma) = \bar{\sigma}\big), \tag{3}$$

and, by the word form $f^*(\tau \cdot \sigma_1 \cdots \sigma_r) = f^*(\tau) \cdot \Psi(\sigma_1 \cdots \sigma_r)$, for a word σ in $\mathcal{E}_f$ one has $\tau \cdot \sigma \in \mathcal{D}_f \iff f^*(\tau) \cdot \Psi(\sigma) = i_*(\tau) \cdot \bar{\sigma}$. In particular the stabilizer of the component through τ is $\{\sigma :\; f^*(\tau)\Psi(\sigma) = i_*(\tau)\bar{\sigma}\}$.

Definition 4.1 (descended twist of a pair)**.** Let δ bound a disc D_δ with $D_\delta \cap B = S$, $|S| \in \{2, 3\}$, and let A_δ be a thin annulus around δ avoiding B. Every component $\mathscr{A} \subset f^{-1}(A_\delta)$ covers A_δ with degree $d \in \{1, 2\}$. Let $h_{\mathscr{A}}$ be the lift to $\mathscr{A}$ of the smallest power that lifts there: a full Dehn twist about the core of $\mathscr{A}$ when $d = 1$; and when $d = 2$ a homeomorphism covering the identity on A_δ whose mapping class on $\mathscr{A}$ is the unique class that covers the full twist on A_δ via the double

cover (constructed explicitly in Theorem 1.4). Extending $h_{\mathscr{A}}$ by the identity off $\mathscr{A}$ and recording its isotopy class on A defines the *descended class* $\rho(\delta) \in \mathrm{MCG}(A)$. Because any two choices for the extension differ by a product of twists about boundary curves that enclose only the unmarked preimages of v (in particular, none of A), the class $\rho(\delta)|_A$ is well-defined; the peripheral ambiguity is invisible on A, and only $\rho(\delta)|_A$ is used below.

Lemma 4.2 (the descended twist of a pair)**.** *For $S = \{p_i, v\}$ a v-pair, the disc D_S contains exactly one critical value (v), $f^{-1}(A_{\delta_S})$ is one degree-2 annulus, the smallest liftable power of T_{δ_S} is $m_0 = 2$, and the branch side U_c satisfies*

$$S = \{p_0, v\} : U_c \cap A = \{p_3\}, \qquad S = \{p_2, v\} : U_c \cap A = \varnothing, \qquad S = \{p_3, v\} : U_c \cap A = \{p_2\}.$$

Hence the descended class on A of the smallest liftable power is trivial for every v-pair: $\Psi(x_i^{\circ 2}) = e$ (the half-lift of Theorem 1.4 applied to the degree-2 annulus). For the cycle pairs $S = \{p_i, p_{i+1}\}$ the preimage has two degree-1 sheets, $m_0 = 1$, and the descended twist is the shifted pair twist $T_{\{p_{i+1}, p_{i+2}\}}$, essential in $\mathrm{PMCG}(A)$.

Proof. The branch-side contents displayed for the v-pairs are computed exactly as in Proposition 3.1 (the model $S = \{p_2, v\}$ computation): the degree-2 disc U_c over D_S contains the full preimage of every point of D_S, namely $c_2 = f^{-1}(v)$ (double) and $\{p_{i-1}, q_{i-1}\} = f^{-1}(p_i)$ (two simple points), and nothing else; the intersection with A is $\{p_{i-1}\}$, with $p_{i-1} = p_3, \varnothing, p_2$ for $i = 0,\ 2$(using Proposition 3.1), 3; the descended class is the twist about the boundary of a disc containing at most one point of A, hence trivial. The cycle-pair case is Lemma 4.3 below. The sheet count is Proposition 3.1's argument: the disc D_S contains $k \in \{0, 1, 2\}$ critical values, and $f^{-1}(D_S)$ has $2 - k$ components, of degrees 1 (if $k = 0$), one degree-2 component (if $k = 1$), or two degree-1 components (if $k = 2$). Apply this to each pair.

$S = \{p_2, v\}$ is done in Proposition 3.1: the lift class is peripheral on A (one disc contains c_2 but *no* point of A; the other contains all of A), hence trivial on A.

For each pair $S = \{p_i, v\}$, the disc D_S contains exactly one of the critical values — $v \in D_S$, with $p_1 \notin D_S$ — so $f^{-1}(D_S) = U_c \ni c_2$ is a single degree-2 disc, $\mathscr{A} = f^{-1}(A_\delta)$ is a single degree-2 annulus, and the smallest liftable power is $m_0 = 2$, realized by the π-rotation of Theorem 1.4. The marked-point content of U_c is determined by the fact that a proper degree-2 disc map contains the full preimage of every point of its image: since $f^{-1}(p_i) = \{p_{i-1}, q_{i-1}\}$ consists of two simple points (indices mod 4; $p_i \neq p_1, v$ is not a critical value), and c_2 already spends the ramification, one has

$$S = \{p_0, v\} :\ U_c \cap A = \{p_3\}; \qquad S = \{p_2, v\} :\ U_c \cap A = \varnothing \text{ (as in Proposition 3.1)}; \qquad S = \{p_3, v\} :\ U_c \cap A = \{p_2\}.$$

In all three cases the branch side contains at most one point of A, so by Definition 4.1 (the peripheral ambiguity is invisible on A) together with Theorem 1.4, the descended twist on A is trivial for the *square*: $\Psi(x_i^2) = e$. This is consistent with the monodromy census (Theorem 6.1), where x_i to the smallest positive power that appears as a monodromy is $x_i^{\circ 2}$ for $i \in \{0, 2, 3\}$ and x_1 for $i = 1$.

$S = \{p_1, v\}$: D_S contains both critical values, so $f^{-1}(A_\delta)$ has two degree-1 components and T_δ lifts directly as a full twist on each. Neither the degrees nor the two-sheeted structure is used in the sequel for this pair: what we need is only the membership $x_1 \in H$, which is the tautological monodromy statement at $\lambda = \frac{1}{2}$ — there the cycle configuration $\{0, \infty, 1, \frac{1}{2}\}$ remains legal and only v loops once around p_1 (simple pole), so that loop lies in M_f and $x_1 \in H$, i.e. $\tau_0 \cdot x_1 \in \mathcal{D}_f$; by Lemma 5.1 the descended class is trivial.

$S = \{p_i, p_{i+1}\}$: D_S contains no critical value, two degree-1 annuli, $m_0 = 1$, and by Lemma 4.3 the descended twist is $T_{\{p_{i+1}, p_{i+2}\}} \neq e$. □

Lemma 4.3 (the shift rule)**.** *For $S \subset A$ a pair with D_S avoiding all critical values, the descended class of T_{δ_S} on A is $T_{\{f(x),f(y)\}}$ where $S = \{x, y\}$; under the 4-cycle action this is $T_{\{p_{i+1},p_{i+2}\}}$ for $S = \{p_i, p_{i+1}\}$.*

Proof. The lift is a full twist on each degree-1 annulus of $f^{-1}(A_\delta)$; on the marked sphere the boundary of the region $f^{-1}(D_S) \ni f^{-1}(S)$ maps to the region $D_S \ni S$, and the induced action on the values $f(S) \subset A$ is the stated pair twist. Essentiality: any pair twist of the 4-punctured sphere is essential in $\mathrm{PMCG}(A) \cong F_2$ (the twists about $\{p_i, p_{i+1}\}$ have nonzero linking number between enclosed and complementary pairs). □

Lemma 4.4 (the descended classes hit all of $\mathrm{PMCG}(A)$)**.** *Let $\mathcal{E}_f = \langle x_1, x_0^{\circ 2}, x_2^{\circ 2}, x_3^{\circ 2}, a_0, a_1, a_2, a_3\rangle$ be as defined at the start of §4, and let $\Psi : \mathcal{E}_f \to \mathrm{PMCG}(A)$ be the descended class, extended to words by the anti-homomorphic rule $\Psi(\sigma_1\sigma_2) = \Psi(\sigma_2)\Psi(\sigma_1)$ (§4). On the generators the descended classes are*

$$\Psi(x_1) = e, \qquad \Psi(x_i^{\circ 2}) = e \ \ (i \neq 1), \qquad \Psi(a_j) = T_{\{p_{j+1},p_{j+2}\}} \ \ (\textit{indices mod } 4),$$

each liftable *(the a_j by Lemma 4.3, x_1 by Theorem 1.4 (iii), $x_i^{\circ 2}$ by Lemma 4.2); the v-pair descended classes are trivial, those of the a_j essential. Here the unsquared x_i ($i \neq 1$) is* not *in $\mathcal{E}_f$ and $\Psi(x_i)$ is not defined; only its square descends (trivially). Consequently $\Psi(\mathcal{E}_f) = \mathrm{PMCG}(A)$.*

Proof. The displayed descended classes are Lemma 4.2 (v-pairs: the smallest liftable power is the square, descending trivially; x_1 lifts directly with trivial descent) and Lemma 4.3 (cycle pairs: $\Psi(a_j) = T_{\{f(p_j),f(p_{j+1})\}} = T_{\{p_{j+1},p_{j+2}\}}$).

On the 4-punctured sphere, a twist about a pair equals the twist about its complement (the two cobound an annulus, hence are isotopic). Therefore $\Psi(a_0) = T_{\{p_1,p_2\}} = T_{\{p_3,p_0\}} = u_0$ and $\Psi(a_2) = T_{\{p_3,p_0\}} = T_{\{p_1,p_2\}} = u_2$, reading off the complementary pair each time; so $\langle \Psi(a_0), \Psi(a_2)\rangle = \langle u_0, u_2\rangle = \mathrm{PMCG}(A)$ (Q1.2).

It remains to show the *set* $\Psi(\mathcal{E}_f)$ equals this group, despite Ψ being an anti-homomorphism. Every element of $\mathcal{E}_f$ is a word w in the generators. By the reversal rule, $\Psi(w)$ is the reversed word in the images: the x_1 and x_i^2 factors contribute e, so $\Psi(w)$ is a word in the $\Psi(a_j)$ with reversed order; each $\Psi(a_j) \in \langle u_0, u_2\rangle$ by the complementary-pair identification above, so $\Psi(w)$ is a word in $u_0, u_2 \in \langle u_0, u_2\rangle$. Since a subgroup is closed under reversal of its words, the reversed word still lies in $\langle u_0, u_2\rangle$; hence $\Psi(w) \in \langle u_0, u_2\rangle$ for every w. Conversely $u_0 = \Psi(a_0), u_2 = \Psi(a_2) \in \Psi(\mathcal{E}_f)$, so $\langle u_0, u_2\rangle \subseteq \Psi(\mathcal{E}_f)$. Hence $\Psi(\mathcal{E}_f) = \langle u_0, u_2\rangle = \mathrm{PMCG}(A)$. □

Proposition 4.5 (content table and its consequence)**.** *For the four pair-twists $x_i = T_{\{p_i,v\}}$ their descended classes on A (Definition 4.1, i.e. the Ψ-value) are:*

where for $i \neq 1$ it is the square *that lifts (the smallest liftable power), each because the relevant branch side contains at most one point of A (Lemma 4.2). By contrast, for the adjacent cycle-pair twists $a_j = T_{\{p_j,p_{j+1}\}}$, the descended class is an essential pair twist $T_{\{p_{i+1},p_{i+2}\}} \neq e$ of* $\mathrm{PMCG}(A)$*. Therefore:*

(a) *$T_\gamma^{\circ m} = x_2^{\circ m}$ preserves $\mathcal{D}_f$ iff m is even (Theorem 1.4).*

(b) *For every other pair-partition curve δ and every $m \neq 0$, the descended class $\Psi(T_\delta^{\circ m})$ is a nonzero power of an essential twist of $(\widehat{\mathbb{C}}, A)$, so $T_\delta^{\circ m}$ does not preserve the component of any point it is applied to; since the descended class depends only on δ (Lemma 4.3, a topological datum, the same at every component), $T_\delta^{\circ m}$ does not preserve $\mathcal{D}_f$ at all.*

(c) *For a non-pair curve δ (enclosing 2 or 3 points — every essential separating curve on the 5-punctured sphere is a pair curve up to isotopy) the same conclusion applies.*

Proof. Part (a) is Theorem 1.4. For (b): the descended-class table is Lemma 4.2; an essential Dehn twist $G_{ij} \in \mathrm{PMCG}(A)$ has infinite order (its action on the Teichmüller space $\mathcal{T}(A)$ — equivalently on the hyperbolic length of the enclosed geodesic — is faithful on length functions; for the 4-punctured sphere $\mathcal{T}(A) \cong \mathbb{H}^2 \times \mathbb{R}$, and G_{ij} acts as a parabolic with a nontrivial translation component). Hence $G_{ij}^m \neq e$ for $m \neq 0$, and $f^*(\tau \cdot T_\delta^{\circ m}) = f^*(\tau) \cdot G_{ij}^m \neq i_*(\tau) = i_*(\tau \cdot T_\delta^{\circ m})$ — the inequality holding on $\mathcal{T}(A)$ because G_{ij}^m acts there without fixed points (the action of $\mathrm{PMCG}(A)$ on $\mathcal{T}(A)$ is free, Q1.1). Applying this at any $\tau \in \mathcal{D}_f$ gives $\tau \cdot T_\delta^{\circ m} \notin \mathcal{D}_f$, so the translated set $T_\delta^{\circ m}(\mathcal{D}_f)$ is disjoint from $\mathcal{D}_f$ near generic points; since both are analytic sets, $T_\delta^{\circ m}(\mathcal{D}_f) \neq \mathcal{D}_f$. Part (c) follows because every essential separating simple closed curve on $\widehat{\mathbb{C}} \setminus B$ ($|B| = 5$) bounds a disc containing exactly 2 or 3 of the points, and the 3-point side is the complement of a pair, so δ is isotopic to a pair curve $T_{\{x,y\}}$ covered by the table. □

5 The covering structure and the component set

We record once, in the generality we use it, the covering fact behind the component set $\pi_0(\mathcal{D}_f)$ sitting inside the sheet space $H\backslash\mathrm{PMCG}(B)$. The decisive step is ((c)): the equalizer is a *right-invariant union of sheets* of $\pi^{-1}(M_f)$.

Lemma 5.1 (equalizer as union of sheets)**.** *Let $M_f \subset \mathcal{M}(B)$ be the matched locus and $\pi : \mathcal{T}(B) \to \mathcal{M}(B)$ the universal cover. The two maps $f^*, i_* : \mathcal{T}(B) \to \mathcal{T}(A)$ both cover maps to $\mathcal{M}(A)$, and on the matched locus they lift one and the same map $M_f \to \mathcal{M}(A)$. Hence:*

(a) *$\mathcal{D}_f \subseteq \pi^{-1}(M_f)$, and on each component S of $\pi^{-1}(M_f)$ the two lifts $f^*|_S, i_*|_S$ either coincide everywhere or nowhere.*

(b) *$\mathcal{D}_f$ is a disjoint union of components of $\pi^{-1}(M_f)$; the covering $\mathcal{D}_f \to M_f$ is unbranched.*

(c) *With base point $\tau_0 = [\mathrm{id}]$ over $* \in M_f$ (which satisfies $\tau_0 \in \mathcal{D}_f$) and $H = \mathrm{image}(\pi_1(M_f, *) \to \mathrm{PMCG}(B))$ (the stabilizer of the sheet through τ_0), the components of $\mathcal{D}_f$ form a* nonempty, right-H-invariant subset *of the sheet space (a union of full H-cosets, containing the identity coset), with the canonical $\mathrm{PMCG}(B)$-equivariant inclusion*

$$\pi_0(\mathcal{D}_f) \;\subseteq\; \pi_0\big(\pi^{-1}(M_f)\big) \;\cong\; H\backslash\mathrm{PMCG}(B), \qquad \text{right translation: } (H\sigma)\cdot g = H(\sigma g).$$

Proof. Both f^*, i_* are lifts of the same map $\nu : M_f \to \mathcal{M}(A)$ (the restriction of the forgetful map $\mathcal{M}(B) \to \mathcal{M}(A)$, with which the pullback correspondence agrees on M_f by the definition of matching). Any two lifts of ν to the covering $\mathcal{T}(A)$ differ by a deck transformation $\delta \in \mathrm{PMCG}(A)$ pointwise constant on each component of $\pi^{-1}(M_f)$ (the difference of two lifts of a connected map to a covering is locally constant and deck-valued). Therefore on each component S either $\delta = e$ ($f^* = i_*$ on S) or $f^*(\tau) = \delta \cdot i_*(\tau) \neq i_*(\tau)$ for all $\tau \in S$ (deck elements act freely). This gives (a) and (b). For (c), the base membership $\tau_0 \in \mathcal{D}_f$ is the tautology at the original map: with $\varphi = \psi = \mathrm{id}$, (1) reads $\mathrm{id} \circ f = f \circ \mathrm{id}$, i.e. $g = f$, so $f^*([\mathrm{id}]) = [\mathrm{id}] = i_*([\mathrm{id}])$. The covering $\pi^{-1}(M_f) \to M_f$ has its sheets permuted transitively by $\mathrm{PMCG}(B)$ with stabilizer H (Q1.1), so its components are the coset space $H\backslash\mathrm{PMCG}(B)$; by ((a)) the truth of "$\in \mathcal{D}_f$" is constant on each sheet, hence the components of $\mathcal{D}_f$ form a *right-H-invariant subset* of $H\backslash\mathrm{PMCG}(B)$ — a union of full H-cosets, nonempty (it contains the identity coset, the sheet through τ_0). Whether every sheet is selected is not determined here and is not needed for the orbit conclusion; what is used is $1 \leq \#\pi_0(\mathcal{D}_f) \leq [\mathrm{PMCG}(B) : H] \leq 4$, the upper bound being Theorem 1.6 (c). □

6 The monodromy of M_f

We now compute the monodromy group H defined by External result 1.1 with $Z = M_f$, proving Theorem 1.6. Fix the base point $* = (2, -\frac{1}{24}) \in M_f$ (Remark 2.2).

Theorem 6.1 (puncture census)**.** *With the loops taken in the parameter sphere $\widehat{\mathbb{C}}_\lambda \setminus X_0$ (Proposition 2.1), the monodromies $\pi_1(M_f, *) \to \mathrm{PMCG}(B)$ of small loops around the punctures are as follows. The list exhausts the ten punctures of X_0 ($2+2+2+1+1+1+1$); since $\pi_1(M_f, *)$ is generated by these loops with the single product-relation, their classes generate the monodromy group H:*

(*i*) *at the two roots $\frac{3\pm\sqrt{5}}{2}$ of $s_0 = \lambda^2 - 3\lambda + 1$: the loop encircles a point where $R(\lambda) = 0$ to order exactly 2 (double contact of v with p_0); the monodromy is $T^{\circ 2}_{\{p_0,v\}} = x_0^{\circ 2}$. (The square, not the twist itself: the derivation is given in the proof below.)*

(*ii*) *at the two roots $s_2 = 0$, i.e. $\lambda_\pm = \frac{-1\pm\sqrt{5}}{2}$: double contact with $\Delta_{\{p_2,v\}}$, monodromy $T^{\circ 2}_\gamma = x_2^{\circ 2}$.*

(*iii*) *at the two roots $s_3 = 0$: double contact with $\Delta_{\{p_3,v\}}$, monodromy $x_3^{\circ 2}$.*

(*iv*) *at $\lambda = \frac{1}{2}$: simple pole of R while the cycle configuration is nondegenerate ($\lambda = \frac{1}{2}$ gives the degenerate map $\beta = 0$, but as a configuration in $\mathcal{M}(A)$ it is legal: $\{0, \infty, 1, \frac{1}{2}\}$ are distinct); the monodromy is the point-push of v once around $p_1 = \infty$, i.e. x_1.*

(*v*) *at $\lambda = 0$: simultaneously $p_3 \to p_0$ and $v \to p_1$ with simple contacts; monodromy $u_0 x_1 = T_{\{p_0,p_3\}} T_{\{p_1,v\}}$ (commuting product, vanishing cycles on different sides of the stratum).*

(*vi*) *at $\lambda = 1$: $p_3 \to p_2$ and $v \to p_1$; monodromy $u_2 x_1 = T_{\{p_2,p_3\}} T_{\{p_1,v\}}$.*

(*vii*) *at $\lambda = \infty$: as $w = \lambda^{-1} \to 0$, both $p_3 = \lambda = w^{-1}$ and $v = R(\lambda) = -\frac{1}{8w}(1 + O(w))$ approach $p_1 = \infty$, with bounded ratio $v/p_3 \to -\frac{1}{8}$, while w itself goes once around 0. The monodromy of the simple w-loop is the commuting product $u_3 x_1$, where $u_3 = T_{\{p_0,p_2\}}$ (the twist about the curve enclosing $\{p_1, p_3, v\}$, equivalently $\{p_0, p_2\}$; it projects to $T_{\{p_1,p_3\}} \in \mathrm{PMCG}(A)$, the push of p_3 around p_1, i.e. $(u_0 u_2)^{-1}$ in the pure group $\mathrm{PMCG}(A) = \langle u_0, u_2 \rangle \cong F_2$, where the loop of p_3 around all three other points is trivial) and $x_1 = T_{\{p_1,v\}}$. In particular $u_3 x_1 \in H$, hence (with $x_1 \in H$ from (iv)) $u_3 \in H$.*

Proof. At each puncture, one chooses a small loop around the puncture in the λ-sphere, lifts the family of 5-pointed spheres $(\widehat{\mathbb{C}}, \{0, \infty, 1, \lambda, R(\lambda)\})$ along it, and computes the monodromy by (Q1.3). For (ii),(iii),(iv) the local equation is

$$R(\lambda) - c = \kappa\, t^2\big(1 + O(t)\big), \qquad t := \lambda - c, \quad \kappa = -\frac{s_i'(c)^2}{D(c)} \neq 0,$$

by (2): the vanishing of $R - c$ is of order *exactly* 2 — its *second* derivative does not vanish — because s_i has a *simple* zero at c (the discriminants $5, 5, -3$ are nonzero, so $s_i'(c) \neq 0$) and $D(c) \neq 0$ (the roots are not in $\{0, \frac{1}{2}, 1\}$ by Proposition 2.1). As $t = e^{i\theta}$, $0 \le \theta \le 2\pi$, the point v therefore circles p_i *twice*, and the loop realizes a full double contact with $\Delta_{\{p_i,v\}}$, giving $T^{\circ 2}_{\{p_i,v\}}$. For (v),(vi): the cycle point $\lambda \to p_i$ collides with contact 1 (the equation $\lambda = c$ itself), while $v = R(\lambda) \to p_1 = \infty$ with contact 1 (the poles of R at $0, 1$ are simple because $D(\lambda)$ has simple zeros there while $s_0(\lambda)^2$ does not vanish: $s_0(0) = 1$, $s_0(1) = -1$); by (Q1.3) the monodromy is the product of the two commuting twists $T_{\{p_i,p_3\}} T_{\{p_1,v\}}$. For (i): R vanishes to order exactly 2 at $c = \frac{3\pm\sqrt{5}}{2}$: writing $t = \lambda - c$ and

$\sigma(t) \coloneqq R(\lambda)/t^2$, one has $\sigma(0) = -s_0'(c)^2/D(c) \neq 0$ (by the discriminant check of Proposition 2.1), so

$$v = R(\lambda) = -s_0'(c)^2\, t^2 \left(1 + O(t)\right) / D(c).$$

Take a loop $\lambda(\theta) = c + \varepsilon e^{i\theta}$, $0 \le \theta \le 2\pi$, with $\varepsilon > 0$ so small that the disc $|z - c| \le \varepsilon$ contains no marked point other than the limiting $p_3 = c$ and that $|v(\theta)| = |s_0'(c)|^2\varepsilon^2(1 + O(\varepsilon))/|D(c)|$ is much smaller than $|c - \{0, 1\}|$ and much smaller than ε — possible since the roots satisfy $c \neq 0, 1$ and $v = O(\varepsilon^2)$ while $p_3 - c = \varepsilon e^{i\theta}$ has modulus ε. Along this loop:

- $p_3(\theta) = c + \varepsilon e^{i\theta}$ winds once around c; the disc bounded by its path contains *no* marked point (the other points $0, 1, \infty$ are at distance $\ge \min(|c|, |1 - c|) \gg \varepsilon$ and $v = O(\varepsilon^2) \ll \varepsilon$ stays off the path), so this motion is isotopic to the identity: the p_3-factor contributes the trivial mapping class.
- $v(\theta) = \kappa\varepsilon^2 e^{2i\theta}(1 + O(\varepsilon))$ winds *twice* around $0 = p_0$ (the argument of v increases by 4π), and the circle $|z| = \kappa\varepsilon^2$ encloses only p_0 among the marked points (p_3 is at distance $\approx |c|$ from 0). The point-push of v once around p_0 is the full twist $x_0 = T_{\{p_0, v\}}$ (Q1.2); two turns give $x_0^{\circ 2}$.

Both motions are simultaneous along the single θ-loop; since the support of the v-push (a small annulus around p_0) is disjoint from the support of the p_3-motion (a small annulus around c), the two factors commute, and the total monodromy is $x_0^{\circ 2} \cdot \mathrm{id} = x_0^{\circ 2}$. This is the monodromy of the simple λ-loop; hence $x_0^{\circ 2} \in H$. For (vii): set $w = \lambda^{-1}$ and take the loop $\lambda(\theta) = \varepsilon^{-1}e^{-i\theta}$, $0 \le \theta \le 2\pi$ (i.e. $w = \varepsilon e^{i\theta}$), with $\varepsilon > 0$ small. From the square identities (2),

$$R(\lambda) = -\frac{\lambda^4(1 + O(w))}{8\lambda^3(1 + O(w))} = -\frac{\lambda}{8}\big(1 + O(w)\big) = -\frac{e^{-i\theta}}{8\varepsilon}\big(1 + O(\varepsilon)\big),$$

so $p_3 = \lambda$ and v both tend to $\infty = p_1$ as $\varepsilon \to 0$, with ratio

$$\frac{v}{p_3} \longrightarrow -\frac{1}{8} \qquad \text{and} \qquad \frac{R(\lambda) - 1}{R(\lambda) - \lambda} = \frac{s_2(\lambda)^2}{s_3(\lambda)^2} \longrightarrow 1 \quad (\text{since } s_2, s_3 \text{ are monic}\ldots \text{ in fact both } \sim \lambda^2).$$

As θ runs from 0 to 2π: $p_3(\theta) = w^{-1}$ winds once around $\infty = p_1$ (one full turn of w around 0), $v(\theta)$ winds once around $\infty = p_1$, and the two are mutually unlinked (their ratio stays in a compact set avoiding $0, \infty$). The marked points 0 and $1 = c = p_0, p_2$ stay bounded, in fact fixed. Choose a small disc Δ around ∞ containing p_1; for ε small, p_3 and v remain inside the annulus $\{C^{-1}\varepsilon^{-1} < |z| < C\varepsilon^{-1}\} \subset \Delta$, while p_0, p_2 remain outside Δ. The motion is the concatenation of: the push of p_3 once around p_1 inside Δ — which on B is the twist about the curve enclosing $\{p_1, p_3\}$, equivalently (by taking the complementary side, p_0, p_2 being fixed outside Δ together with v at the moment of pushing p_3) the class $u_3 = T_{\{p_0, p_2\}}$, projected in $\mathrm{PMCG}(A)$ to $T_{\{p_1, p_3\}}$ — and the push of v once around p_1, which is x_1. Since the two pushes are realized on disjoint annuli (p_3 moving at radius $|w|^{-1}$, v at radius $|8w|^{-1}$, separated by the fixed ratio), they commute, and the monodromy is u_3x_1. Hence $u_3x_1 \in H$, and with $x_1 \in H$ from (iv), also $u_3 = (u_3x_1)\, x_1^{-1} \in H$ and $u_3^{\circ 2} \in H$. □

Corollary 6.2 (structural form of H)**.** *Let $K = \langle x_0, x_1, x_2, x_3 \mid x_0x_1x_2x_3 = 1\rangle \cong F_3$ be the Birman kernel (Q1.2), free on the* consecutive triple $\{x_0, x_1, x_2\}$ *with $x_3 = (x_0x_1x_2)^{-1}$. For $i = 0, 1, 2$ let $\mathrm{ch}_i : K \to \mathbb{Z}$ be the exponent-sum homomorphism in x_i on this free basis (so $\mathrm{ch}_i(x_j) = \delta_{ij}$ for $i, j \in \{0, 1, 2\}$, and ch_i is the exponent-sum of x_i in every word; it is also the linking number of the v-strand with the p_i-strand of the corresponding pure braid). The monodromy group H (Theorem 6.1) satisfies:*

(*i*) *the following elements lie in* $H \cap K$*:*

$$x_1 \text{ (census (iv))}, \qquad x_0^{\circ 2} \text{ ((i))}, \qquad x_2^{\circ 2} \text{ ((ii))}, \qquad (x_0x_1x_2)^{-2} = x_3^{\circ 2} \text{ ((iii))};$$

(*ii*) $H \to \mathrm{PMCG}(A) \cong F_2$ *is onto* (*images of* u_0x_1, u_2x_1 *are* u_0, u_2) *with kernel* $H \cap K$*;*

(*iii*) $H \cap K \subseteq \mathrm{ch}_0^{-1}(2\mathbb{Z}) \cap \mathrm{ch}_2^{-1}(2\mathbb{Z}) =: F$*: each of the ten puncture-loop monodromies of Theorem 6.1 has even* ch_0 and *even* ch_2 — x_1 *and* u_0x_1, u_2x_1, u_3x_1 *have* $(0,0)$*;* x_0^2 *has* $(2,0)$*;* x_2^2 *has* $(0,2)$*;* $(x_0x_1x_2)^{-2}$ *has* $(-2,-2)$ — *and* $(\mathrm{ch}_0, \mathrm{ch}_2) : K \to \mathbb{Z}^2$ *is a homomorphism. Since every element of* H *is a word in these loop classes* (*the loops generate* $\pi_1(M_f)$)*, every element of* $H \cap K$ *has even charges. The subgroup* F *has index* 4 *in* $K = \langle x_0, x_1, x_2 \rangle = F_3$*;*

(*iv*) $x_2 \notin H$ ($\mathrm{ch}_2(x_2) = 1$ *is odd, so (iii) excludes it from* $H \cap K$*, and* $x_2 \in K$) *while* $x_2^{\circ 2} \in H$*.*

Consequently

$$\left\langle x_1,\, x_0^{\circ 2},\, x_2^{\circ 2},\, (x_0x_1x_2)^{-2} \right\rangle^H \;\leq\; H \cap K \;\leq\; F, \qquad H/(H \cap K) \xrightarrow{\sim} F_2,$$

and $2 \leq [\mathrm{PMCG}(B) : H] = [K : H \cap K] \leq 4$*, with the lower bound from* (*iv*)*.*

Proof. Items (i) and (ii) read off items (i)—(vi) of Theorem 6.1 together with (Q1.2). For (iii), let

$$H_0 \;\coloneqq\; \left\langle\, x_1,\; x_0^2,\; x_2^2,\; (x_0x_1x_2)^{-2},\; u_0,\; u_2,\; u_3 \,\right\rangle \;\subseteq\; H :$$

indeed the census supplies the four K-elements and the products u_ix_1 ($i \in \{0,2\}$ from (v),(vi), $i = 3$ from (vii)); since $x_1 \in H$ by (iv) of the census, these give $u_i = (u_ix_1)\, x_1^{-1} \in H$ for $i \in \{0,2,3\}$, and conversely the u_i recover the u_ix_1. Because the census loops exhaust the ten punctures of M_f, $\pi_1(M_f)$ is generated by their classes (with the single product-relation), so $H = H_0$; the displayed decomposition is legitimate. The projection of H_0 to $\mathrm{PMCG}(A)$ is generated by u_0, u_2, u_3, hence (using $T_{\{p_1,p_3\}} = (u_0u_2)^{-1}$ of the census item (vii)) by u_0, u_2, and therefore surjects onto $\mathrm{PMCG}(A) = \langle u_0, u_2 \rangle$; its kernel is $H_0 \cap K$. The seven displayed generators each have even $(\mathrm{ch}_0, \mathrm{ch}_2)$-value as tabulated in the statement — the u_j's, whose defining curves avoid v, have value $(0,0)$. Since $(\mathrm{ch}_0, \mathrm{ch}_2) : K \to \mathbb{Z}^2$ is a *homomorphism*, its value on any word in these generators and their inverses is the sum of the values; as every census element lies in H_0, every element of $H \cap K$ arises as such a word projected trivially, hence $H \cap K \subseteq F$. Finally (iv) is immediate:

$\mathrm{ch}_2(x_2) = 1 \notin 2\mathbb{Z}$, so $x_2 \notin F \supseteq H \cap K \ni x_2$ would contradict; more precisely $x_2 \notin H$ because were $x_2 \in H$ then $x_2 \in H \cap K \subseteq F$, contradiction. And $x_2^{\circ 2}$ is a census generator. The bounds $2 \leq [\mathrm{PMCG}(B) : H] \leq 4$ follow from $[K : F] = 4$ and $x_2 \notin H \cap K$. □

Corollary 6.3 (index and components)**.** $\pi_0(\mathcal{D}_f)$ *is a nonempty subset of the finite coset space* $H\backslash\mathrm{PMCG}(B)$*, with* $2 \leq [\mathrm{PMCG}(B) : H] \leq 4$*, hence* $1 \leq \#\pi_0(\mathcal{D}_f) \leq 4$*; in fact* $\#\pi_0(\mathcal{D}_f) \in \{2,4\}$ *is even. The twist* $T_\gamma^{\circ 2} = x_2^{\circ 2} \in H$ *acts as the identity on* $H\backslash\mathrm{PMCG}(B)$*, hence on* $\pi_0(\mathcal{D}_f)$*; the odd twist* $T_\gamma = x_2 \notin H$ *acts with orbits of size exactly* 2 (*no fixed component*)*.*

Proof. By Lemma 5.1, $\pi_0(\mathcal{D}_f) \subseteq H\backslash\mathrm{PMCG}(B)$ is a nonempty, right-H-invariant subset (a union of full cosets). By Corollary 6.2 (iii), $H \cap K \subseteq F$ with $[K : F] = 4$, and H surjects onto $\mathrm{PMCG}(A)$ with kernel $H \cap K$ (Corollary 6.2 (ii)); hence

$$[\mathrm{PMCG}(B) : H] = [K : H \cap K] \leq [K : F] = 4,$$

and this is ≥ 2 because $x_2 \in K \setminus H$ (Corollary 6.2 (iv)). The action of $\mathrm{PMCG}(B)$ on $H\backslash\mathrm{PMCG}(B)$ is by right translation, so any element of H — in particular $x_2^{\circ 2}$, which lies in H by Corollary 6.2 (i)

— fixes every coset, hence acts as the identity on $H\backslash\mathrm{PMCG}(B)$ and on its subset $\pi_0(\mathcal{D}_f)$. For the odd twist: right multiplication by x_2 permutes the *selected* components, because $\tau_0 \cdot x_2 \in \mathcal{D}_f$ — the monodromy loop at the double zero of s_2 has winding number 2, so the two lifts of a parametrized small loop in M_f starting at τ_0 and at $\tau_0 \cdot x_2$ agree, whence the sheets through τ_0 and $\tau_0 \cdot x_2$ are simultaneously selected (part ((a)) of Lemma 5.1); and the selected set is right-H-invariant, so x_2 takes the identity coset $H \in \pi_0(\mathcal{D}_f)$ to $Hx_2 \in \pi_0(\mathcal{D}_f)$, and by H-invariance and $x_2^2 \in H$ it pairs up the whole selected set. It has no fixed component: $Hg = Hgx_2$ would give $gx_2g^{-1} \in H \cap K$ (as K is normal), but ch_2 is conjugacy-invariant on $\mathrm{PMCG}(B)$ (pure braids fix the labels v, p_2), so $\mathrm{ch}_2(gx_2g^{-1}) = \mathrm{ch}_2(x_2) = 1$, odd, hence outside $F \supseteq H \cap K$ (Corollary 6.2 (iii)). Thus the selected components pair up under x_2, so $\#\pi_0(\mathcal{D}_f)$ is even, in $\{2, 4\}$. □

Proof of Theorem 1.7. The component count is Corollary 6.3; the actions there and Proposition 4.5 (b)–(c) give the twist statements. □

Proof of Corollary 1.8. Any power of a Dehn twist preserving $\mathcal{D}_f$ must, by Proposition 4.5, be an even power of T_γ; the action of $T_\gamma^{\circ 2}$ on $\pi_0(\mathcal{D}_f)$ is the identity (and that of T_γ is an involution with 2-cycles, on a set of at most 4 elements). Hence no infinite orbit exists. The ultimate reason is the perfect-square structure of Theorem 1.5, which forces $x_2^{\circ 2} = T_\gamma^{\circ 2}$ to occur as monodromy of the matched locus (Theorem 6.1(ii)), i.e. the locus itself knows how to loop around its own degenerations. □